

Section 1.3 Hw. Write-up

16. Find a sentence that is logically equivalent to $(\neg p) \rightarrow q$ using only $\neg$ and $\wedge$.

$$\begin{aligned}
 (\neg p) \rightarrow q &\equiv \neg(\neg p) \vee q \text{ by implication expansion} \\
 &\equiv p \vee q \text{ by double negation} \\
 &\equiv \neg(\neg p \wedge (\neg q)) \text{ by De Morgan's Law}
 \end{aligned}$$

p	q	$(\neg p) \rightarrow q$	$(\neg p) \wedge (\neg q)$	$\neg(\neg p \wedge (\neg q))$
T	T	T	F	T
T	F	T	F	T
F	T	T	F	T
F	F	F	T	F

The truth columns are the same so

$(\neg p) \rightarrow q$ is logically equivalent to $\neg(\neg p \wedge (\neg q))$

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p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	F
F	F	T	T
F	F	F	F

$$? = (p \wedge q \wedge (\neg r)) \vee (p \wedge (\neg q) \wedge r) \vee (p \wedge (\neg q) \wedge (\neg r)) \vee (\neg p \wedge (\neg q) \wedge r)$$

69. What are the antecedent, consequent, and contrapositive of $n^2 > 4$ when $n > 2$?

Rewrite if $n > 2$ then $n^2 > 4$

Antecedent is $n > 2$, Consequent is $n^2 > 4$. Contrapositive is

If n^2 is not greater than 4 then n is not greater than 2